Math 1220 Summer
Exam 2 Practice

Name: ______________________

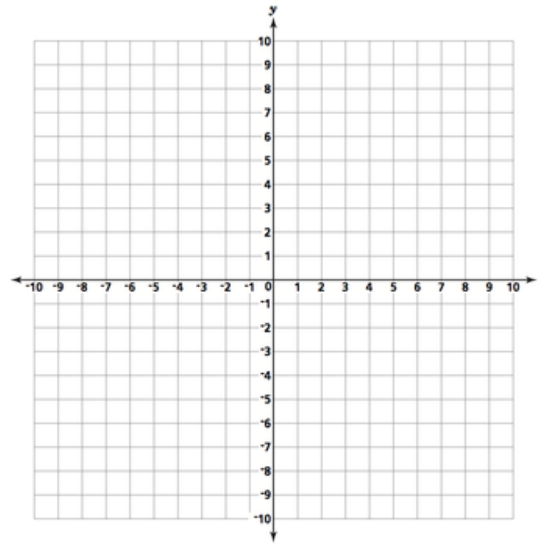
Follow all written homework guidelines. Use proper notation. Show work clearly. Answers must be justified by your work.

1. *(14 pts)* Find the $x$ and $y$ intercepts of $p(x) = \frac{1}{4}(x+1)^5 - 8$, determine the transformation, and then graph the function.

x-int: ______________

y-int: ______________

Transformation:



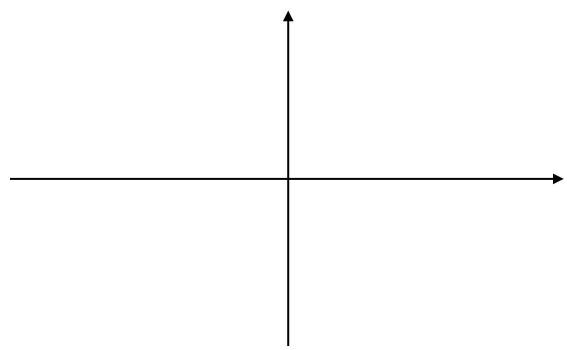

2. *(12 pts)* Let $h(x) = -x(x+1)^2(x-3)^3$

(a) Determine the leading term: ___________

(b) Write the value and multiplicity of each real zero (x-intercept)

| Value | Multiplicity |
| --- | --- |
| | |

(c) Graph $h(x)$. Mark the real zeros as numbers on the axis.



3. *(14 pts)* Let $g(x) = x^4 - 5x^3 - 6x^2$

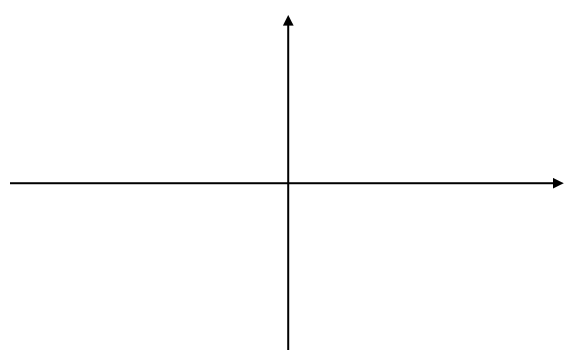
(a) Determine the leading term: ___________

(b) Factor g(x) completely

(c) Write the value and multiplicity of each real zero (x-intercept)

| Value | Multiplicity |
| --- | --- |
| | |

(d) Graph $h(x)$. Mark the real zeros as numbers on the axis.



4. *(10 pts)* Find the equation of a function whose vertex is (1, -4) and passes through the points (2,-2).

5. *(20 pts)* Let $f(x) = 2x^2 - 4x - 6$

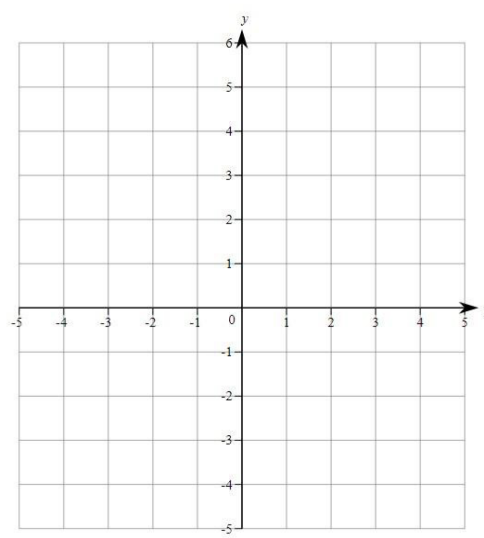
(a) Write *f(x)* in vertex form

(d) Find the *x*-intercepts.

(b) Vertex: ________________

(c) Find the *y*-intercept.

(e) Graph the function. Make sure to indicate at least 5 points and the symmetry axis.



6. *(18 pts)* Find all the asymptotes, delete points that exist, graph and state the domain and range for the function below.

$$f(x) = \frac{x-2}{x^2-3x+2}$$

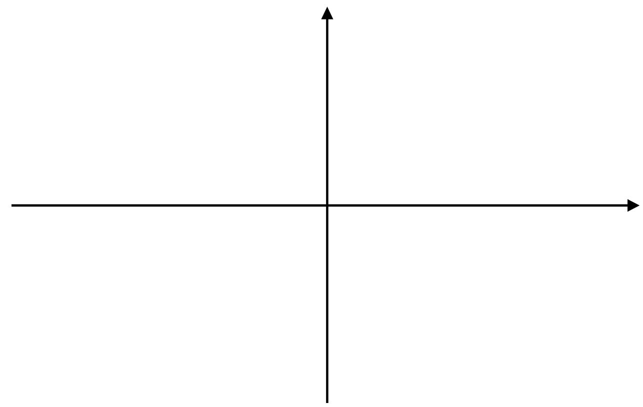
Vertical Asymptote: ______________

Horizontal Asymptote: ____________

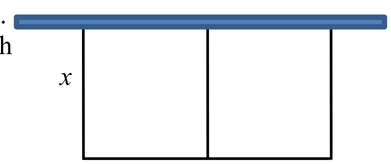
Deleted Point: __________________

Domain: __________________

Range: ___________________



7. *(12 pts)* A gardener has 100 feet of fencing to enclose three adjacent rectangular growing plots. An existing fence borders the north side, so no fencing is needed there. Let $x$ represent the width of the plot as shown. **Note:** Be sure to label any variables you use in the picture.




(a) Express the enclosed area, $A$, in terms of $x$.

(b) Determine the dimensions that will maximize the enclosed area.

8 ) $g(x) = \frac{(x-2)^2}{(x+1)(x-3)}$

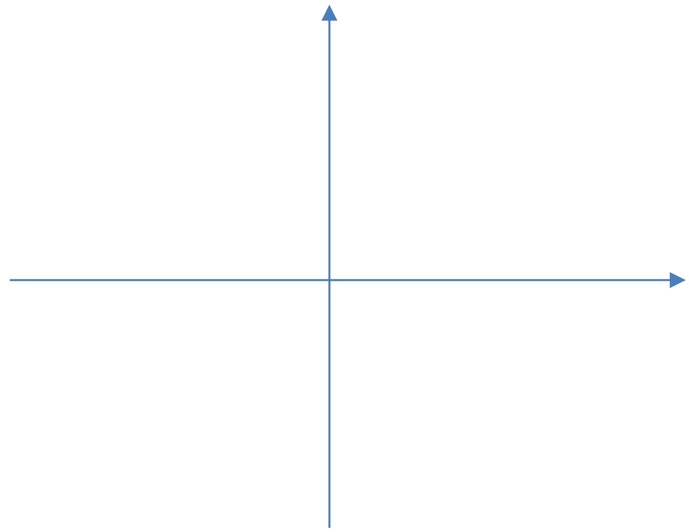
VA(s): ______________

x-int: _______________

HA(s): ______________

y-int: _______________

DP: ________________



| | |
|---|---|
| $x$-intervals: | |
| Above or Below: | |
| Justification: | |

9. Convert each from exponential to logarithmic form, or vice-versa.

(a) $3^x = y$  (b) $e^x = 8$  (c) $\log_3 9 = 2$  (d) $\log x = 6$

(e) $\mathrm{A} = \mathrm{B}^{\mathrm{C}}$  (f) $2^{x-1} = y$  (g) $\log_x \mathrm{Y} = \mathrm{Z}$  (h) $\ln(x-1) = y + 2$

10. Evaluate by "talking it out." (a) $\log_2 16$: "the ____________________on base _____which gives ______" = _______

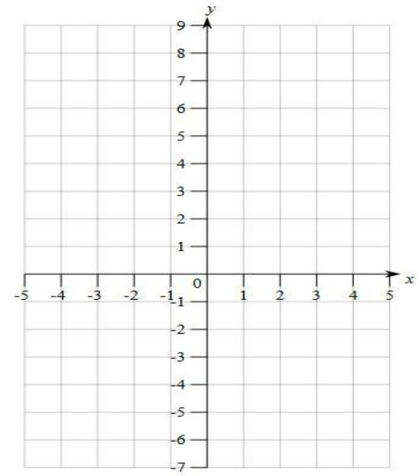
11. Simplify (a) $\log_2 32 =$ _____  (b) $\log_9 3 =$ _____  (c) $\log_8 \frac{1}{8} =$ _____  (d) $\log_6 \sqrt{6} =$ _____

(e) $\log_\pi 1 =$ _____  (f) $\log_p p =$ _____  (g) $\log_{27} 3 =$ _____  (h) $\ln(\frac{1}{e^2}) =$ _____

12. Evaluate (a) $\log_3 3^8 =$ _____  (b) $(\log_3 3)^8 =$ _____  (c) $2 \ln e^5 =$ _____  (d) $10^{\log 3\pi} =$ _____

(e) $(\log_4 2)^2 =$ _____  (f) $2e^{3 \ln 2} =$ _____  (g) $\log(\ln e^{100}) =$ _____  (h) $\log_3\left(10^{2 \log 3}\right) =$ _____

13. Graph function $f(x) = \log_2(x+1) - 1$ by finding x-intercept and label asymptotes.




14.Separate Logarithm $\quad \mathbf{\log_3 \frac{x^3 y}{27z^2}}$

15. **True or False:** If false, change the left side of the statement (in the most appropriate way) to make it true.

(a) _____ $(\ln x)^2 = 2\ln x$

(b) _____ $3e^{\ln x} = e^{\ln x^3}$

(c) ______ $\log 10^3 = 1^3$

(d) _____ $3e^{\ln(x+1)} = 3(x+1)$

(e) _____ $\log_2 3x^4 = 4\log_2 3x$

(f) _____ $\ln(8x^3) = 3\ln 2x$

(g) _____ $\frac{\ln 6x}{2} = \ln 3x$

(h) _____ $\frac{1}{\log 2} = \log_2 10$

(i) _______ $\frac{\log_2 9}{\log_2 5} = \log_2 9 - \log_2 5$

(j)____$\log_5\left(\frac{a}{b^2}\right) = \log_5 a - 2\log_5 b$

(j) ____ $\log(x-y) = \log x - \log y$

(k) _____ $\log(x+y) = \log x + \log y$

(l) _____ $\frac{logx}{logy} = \log\frac{x}{y}$

(m) _____ $\ln A = -\ln\left(\frac{1}{A}\right)$

(n) ____ $(\log_2 7)^x = x\log_2 7$

(o) ____ $\log P \cdot \log Q = \log PQ$

16. Solve for x

(a) $4e^{2x-5} + 3 = 11$

(b) $3^{2-3x} = 8$

(c) $3^x = e^{x+1}$

(d) $3^{6x+5} = 5^{2x}$

10. Solve for x.

(a) $\ln 4x = 2$

(b) $2\log_2 4x - 3 = 5$

(c) $2 \ln \sqrt{x-1} = \ln(x+3) - \ln x$

(d) $\log_4(x-2) = 1 - \log_4(x-5)$